

20) Suppose A, B and X are $n \times n$ matrices with A, X , and $A-AX$ invertible, and suppose

$$(A-AX)^{-1} = X^{-1}B \quad (3)$$

a) Explain why B is invertible.

Solution:

By the definition of invertible matrices, there exists unique matrices A^{-1}, X^{-1} and $(A-AX)^{-1}$ s.t. $A^{-1}A = AA^{-1} = I$, $X^{-1}X = XX^{-1} = I$, $(A-AX)^{-1}(A-AX) = (A-AX)(A-AX)^{-1} = I$.

$$(A-AX)^{-1} = X^{-1}B \Rightarrow X(A-AX)^{-1} = X(X^{-1}B) \Rightarrow X(A-AX)^{-1} = (XX^{-1})B \quad (\text{Theorem 2(a)})$$

$$\Rightarrow X(A-AX)^{-1} = IB \quad (\text{By the definition of inverse})$$

$$\Rightarrow X(A-AX)^{-1} = B \quad (\text{Theorem 2(e)})$$

By Theorem 6(a), we know $(A-AX)^{-1}$ is invertible, and $((A-AX)^{-1})^{-1} = (A-AX)$. By Theorem 6(b), $X(A-AX)^{-1}$ is invertible and $B^{-1} = (X(A-AX)^{-1})^{-1} = ((A-AX)^{-1})^{-1}X^{-1} = (A-AX)X^{-1}$. $\therefore B$ is invertible.

b) Solve (3) for X . If you need to invert a matrix, explain why that matrix is invertible.

Solution:

$$(A-AX)^{-1}(A-AX) = I$$

$$(A-I-AX)(X^{-1}B) = I \Rightarrow (A-AX)(X^{-1}B) = I \Rightarrow$$

$$\Rightarrow A(I-X)(X^{-1}B) = I \quad (\text{Theorem 2(e)})$$

$$\Rightarrow A(I-X)(X^{-1}B) = I \quad (\text{Theorem 2(b)})$$

$$\Rightarrow A((I-X)(X^{-1}B)) = I \quad (\text{Theorem 2(a)})$$

$$\Rightarrow A^{-1}(A((I-X)(X^{-1}B))) = A^{-1}I$$

$$\Rightarrow (A^{-1}A)((I-X)(X^{-1}B)) = A^{-1}I \quad (\text{Theorem 2(a)})$$

$$\Rightarrow I((I-X)(X^{-1}B)) = A^{-1}I \quad (\text{Defn. of inverse})$$

$$\Rightarrow (I-X)(X^{-1}B) = A^{-1}I \quad (\text{Theorem 2(e)})$$

$$\Rightarrow (I-X)(X^{-1}B) = A^{-1}I \quad (\text{Theorem 2(a)})$$

$$\Rightarrow (I-X)X^{-1}B = A^{-1}I \quad (\text{Theorem 2(c)})$$

$$\Rightarrow (IX^{-1} - XX^{-1})B = A^{-1}I \quad (\text{Theorem 2(e)})$$

$$\Rightarrow (X^{-1} - XX^{-1})B = A^{-1}I \quad (\text{By the defn. of inverse})$$

$$\Rightarrow (X^{-1} - I)B = A^{-1}I$$